AI INTERNSHIP • MODULE 01

Learning Notes

Introduction to AI & Python

Day 1 – Day 4 | Beginner Level

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MODULE OBJECTIVES

1. Learn the basics of AI and Machine Learning 2. Understand AI vs ML vs Deep Learning 3. Explore real-world AI applications 4. Install Python and practise core programming concepts 5. Write beginner-friendly Python programs

PART 01

Basics of AI & Machine Learning

Foundations you need before writing a single model

Artificial Intelligence (AI) is the field of building computer systems that perform tasks normally requiring human intelligence: understanding language, recognising images, making decisions and solving problems.

Machine Learning (ML) is a subset of AI where systems **learn patterns from data** instead of being given explicit rules. The more good-quality data a model sees, the better it tends to perform.

KEY TAKEAWAY
Traditional programming: **Rules + Data** → **Answers**. Machine learning: **Data + Answers** → **Rules (the model)**.

Types of Machine Learning

Type	How it learns	Example
Supervised	Labelled data: inputs paired with correct answers	Spam detection, price prediction
Unsupervised	Finds hidden structure in unlabelled data	Customer segmentation
Reinforcement	Trial and error, guided by rewards	Game agents, robotics

The ML workflow

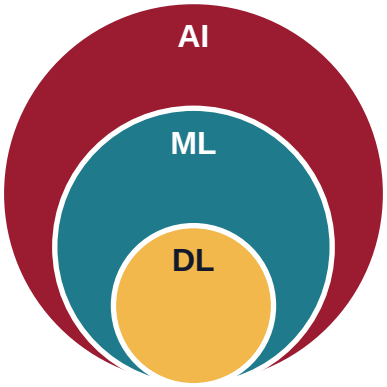
PRO TIP
Always evaluate on data the model has **never seen** (a test set). A model that only memorises its training data is overfitting and will fail in the real world.

PART 02

AI vs ML vs Deep Learning

Three terms, one nested family

The three terms are nested like Russian dolls: Deep Learning sits inside Machine Learning, which sits inside AI.



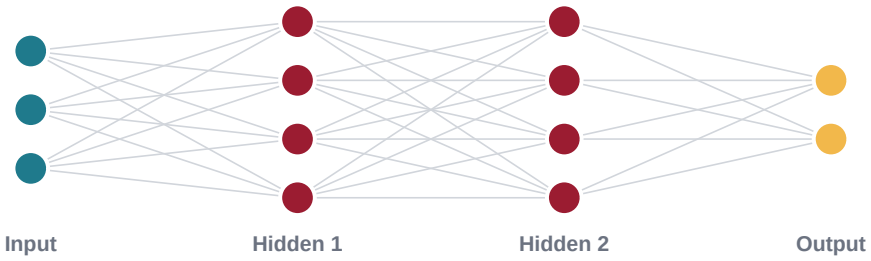
AI Artificial Intelligence: machines that mimic intelligent behaviour

ML Machine Learning: systems that learn patterns from data

DL Deep Learning: multi-layer neural networks

	AI	Machine Learning	Deep Learning
Scope	Any intelligent machine behaviour	Learns from data	Many-layered neural networks
Data need	Varies; rule-based needs little	Moderate	Very large
Features	Hand-written rules	Often engineered by hand	Learned automatically
Hardware	Standard	CPU is usually enough	Typically GPUs
Examples	Chess engines, expert systems	Decision trees, regression	Image CNNs, transformers

Inside a neural network



KEY TAKEAWAY

Each hidden layer transforms the data a little further, so the network can build up from simple patterns (edges) to complex ones (faces, objects).

PART 03

Real-World AI Applications

AI is already part of everyday life

Domain	Applications
Healthcare	Medical image analysis, drug discovery, patient risk prediction
Finance	Fraud detection, credit scoring, algorithmic trading
Retail	Recommendations, demand forecasting, support chatbots
Transport	Self-driving vehicles, route optimisation, traffic prediction
Education	Personalised learning, automated grading, AI tutors
Agriculture	Crop disease detection, yield prediction, precision farming
Entertainment	Content recommendations, game AI
Daily life	Voice assistants, spam filters, translation, face unlock

Responsible AI

- **Data quality:** biased data leads to biased results.
- **Oversight:** AI supports human decisions and can still make mistakes.
- **Ethics:** privacy, fairness and transparency are central concerns.

PART 04

Python Setup & Core Concepts

From installation to your first functions

Installing Python

- Download Python 3 from **python.org**.
- On Windows, tick **Add Python to PATH** during setup.
- Use VS Code or PyCharm as your editor.
- Install libraries with `pip install name`; isolate projects with `python -m venv venv`.

TERMINAL: VERIFY INSTALL

```
python --version
pip --version
```

Variables & data types

PYTHON

```
name = "Asha"        # str
age = 21             # int
height = 1.62        # float
is_student = True    # bool
print(type(age))     # <class 'int'>
```

Operators & input

PYTHON

```
a, b = 10, 3
print(a + b, a - b, a * b)    # 13 7 30
print(a / b, a // b, a % b)   # 3.33.. 3 1
print(a ** b)                 # 1000

user = input("Enter your name: ")
print(f"Hello, {user}!")
```

Conditionals & loops

PYTHON

```
marks = 72
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Keep practising")

for i in range(1, 6):
    print(i)

count = 3
while count > 0:
    print(count)
    count -= 1
```

Functions

PYTHON

```
def square(n):
    return n * n

def greet(name="friend"):
    return f"Hello, {name}"

print(square(5))   # 25
print(greet())     # Hello, friend
```

Lists

PYTHON

```
fruits = ["apple", "banana", "cherry"]
fruits.append("mango")
fruits.remove("banana")
print(fruits[0], len(fruits))
print(fruits[-1])                    # last item
print([f.upper() for f in fruits])   # list comprehension
```

Method	Purpose
append(x)	Add an item at the end
insert(i, x)	Add an item at index i
remove(x)	Remove the first occurrence of x
pop()	Remove and return the last item
sort()	Sort the list in place
len(list)	Number of items

PART 05

Beginner Python Programs

Six small programs to practise the fundamentals

1. Even or odd

PYTHON

```
n = int(input("Enter a number: "))
print("Even" if n % 2 == 0 else "Odd")
```

2. Simple calculator

PYTHON

```
def calc(a, b, op):
    if op == "+": return a + b
    if op == "-": return a - b
    if op == "*": return a * b
    if op == "/": return a / b if b != 0 else "Cannot divide by zero"
    return "Invalid operator"

print(calc(8, 2, "/"))   # 4.0
```

3. Factorial

PYTHON

```
def factorial(n):
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result

print(factorial(5))   # 120
```

4. Fibonacci series

PYTHON

```
a, b = 0, 1
for _ in range(10):
    print(a, end=" ")
    a, b = b, a + b
```

5. Largest number in a list

PYTHON

```
nums = [14, 7, 92, 33, 5]
largest = nums[0]
for n in nums:
    if n > largest:
        largest = n
print("Largest:", largest)   # 92
```

6. Palindrome check

PYTHON

```
word = input("Word: ").lower()
print("Palindrome" if word == word[::-1] else "Not a palindrome")
```

PART 06

Summary & Revision Checklist

Everything from Module 01 at a glance

- ✓ AI is the goal, ML learns from data, Deep Learning uses deep neural networks.
- ✓ Three ML types: supervised, unsupervised, reinforcement.
- ✓ AI is applied across healthcare, finance, retail, transport, education and more.
- ✓ Python installed and verified with python --version.
- ✓ Core Python: variables, operators, conditionals, loops, functions, lists.
- ✓ Practised: even/odd, calculator, factorial, Fibonacci, largest number, palindrome.

DELIVERABLES REQUIRED

- GitHub repository with your Python practice programs
- Learning notes (this document)

Modules 02 onwards were locked when these notes were prepared and can be appended once unlocked.